

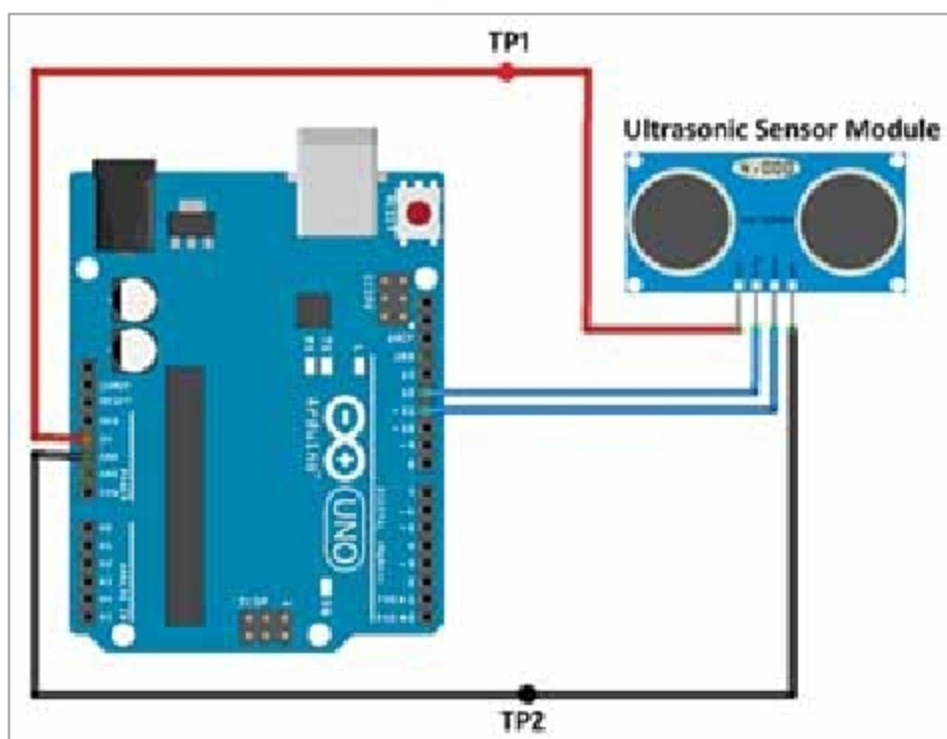


Fig. 4: Connections diagram for the project

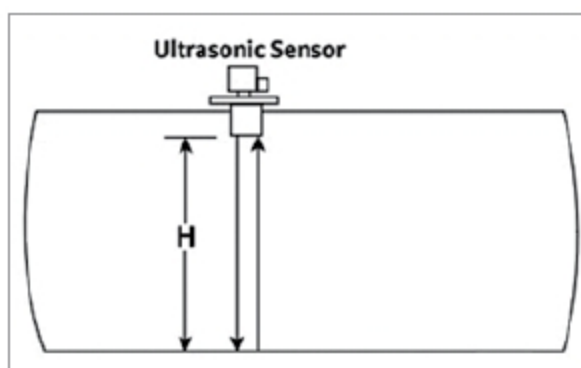


Fig. 5: Empty tank

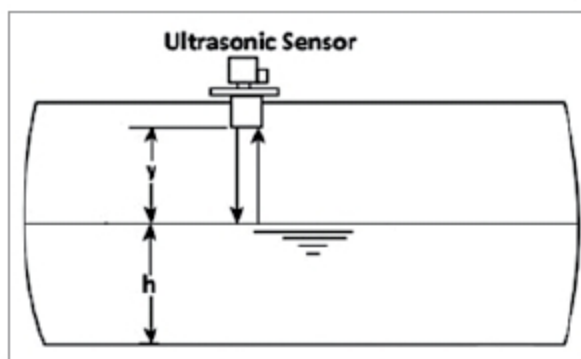


Fig. 6: Tank filled with liquid

The connections diagram for the project is shown in Fig. 4.

The ultrasonic sensor measures the distance between the sensor and object/liquid by sending an ultrasonic wave and calculating the time required for the echo to return. If the time between the trigger and echo is known, the distance of the object can be calculated by simply multiplying half of the time with the speed of sound in air. This principle is used here to detect the level of liquid in a tank. The working of the sensor is illustrated in Figs 5 and 6.

As shown in Fig. 5, when the tank is empty, the ultrasonic wave hits the bottom of the tank and gets

reflected. As a result, it detects the depth of the empty tank at 'H' height. If the tank is filled with liquid up to 'h' height, the ultrasonic wave is reflected from the liquid's top surface, and it detects the depth of empty space 'y' above the liquid, as shown in Fig. 6. When 'H' and 'y' are known,

the liquid level 'h' can be calculated by subtracting 'y' from 'H,' that is, $h = (H - y)$. The value of H is calculated only once during the initial calibration when the tank is empty. For our setup, the value of H was 17cm.

Software

The GUI application program has been developed in the R2014a version of MATLAB. For the Arduino code, the 'NewPing' library has been used. The steps for programming Arduino board are:

1. Open Arduino IDE.
2. Go to Sketch→Include Library→Add .ZIP Library.
3. Browse and select NewPing_v1.7. (The library is provided with the source code.)
4. Connect the Arduino board and select the proper board and COM port from the Tools menu.
5. Upload the code 'Arduino_code.ino' to the board.

The MATLAB code consists of files 'code.m' and 'code.fig'. Keep both the files in the same folder. Open the file 'code.m.' Find the line `s = serial('COM9')` (on line no. 88), and replace 'COM9' with COM port on which your Arduino board is installed.

Run the file 'code.m.' After running this file, GUI will appear, as shown in



Fig. 7: Calibrated water container

Test Points for Troubleshooting

Test point	Expected voltage
TP1	5V
TP2	GND (0V)

Fig. 2. Click on the 'Connect' button to connect Arduino with MATLAB. Click on the 'Run' button to monitor the level of liquid. Click on the 'Stop' button to stop monitoring. To disconnect MATLAB and Arduino, click on the 'Disconnect' button.

Construction and testing

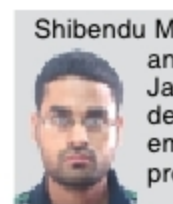
Make the connections as shown in Fig. 4. Place the ultrasonic sensor at top of the tank.

You can test the project using a suitable plastic water container. The ultrasonic module can be fixed on top of the container, as shown in Fig. 1. If the reading shows 0, it means the tank is empty. When the water level rises, readings will also keep increasing. If the reading shows that the water is up to the maximum level, it's time to close the water inlet of the tank (or stop the water pump).

The voltage levels at different test points in the connections diagram are shown in the test point table. **EFY**



Saikat Patra is passionate about electronics and MCU based embedded system applications



Shibendu Mahata is M.Tech in Instrumentation and Electronics Engineering from Jadavpur University. He loves designing MCU based real-time embedded signal processing and process control systems

EFY Note

The source code of this project is available for free download at source.efymag.com